

Mucilage exudation facilitates root water uptake in dry soils

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Abstract. As plant roots take up water and the soil dries, water depletion is expected to occur in the rhizosphere. However, recent experiments showed that the rhizosphere was wetter than the bulk soil during root water uptake. We hypothesise that the increased water content in the rhizosphere was caused by mucilage exuded by roots. It is probably that the higher water content in the rhizosphere results in higher hydraulic conductivity of the root–soil interface. In this case, mucilage exudation would favour the uptake of water in dry soils. To test this hypothesis, we covered a suction cup, referred to as an artificial root, with mucilage. We placed it in soil with a water content of $0.03 \text{ cm}^3 \text{ cm}^{-3}$, and used the root pressure probe technique to measure the hydraulic conductivity of the root–soil continuum. The results were compared with measurements with roots not covered with mucilage. The root pressure relaxation curves were fitted with a model of root water uptake including rhizosphere dynamics. The results demonstrated that when mucilage is added to the root surface, it keeps the soil near the roots wet and hydraulically well conductive, facilitating the water flow from dry soils towards the root surface. Mucilage exudation seems to be an optimal plant trait that favours the capture of water when water is scarce.

Additional keywords: artificial root, chia seeds, rhizosphere, root pressure probe, *Salvia hispanica*.

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Introduction

In the coming decades, water is expected to become a major constraint for agriculture worldwide (Food and Agriculture Organization 2012). Additionally, precipitation is predicted to become sparser and more irregular, increasing the risks of drought spells and yield losses in large areas of our globe. It is therefore of high interest to better understand the mechanisms that help plants to tolerate drought better.

According to Blum (2005), plants can resist drought either by dehydration avoidance or by dehydration tolerance. To limit the risk of dehydration, plants can reduce transpiration by closing stomata or they can grow roots that are more efficient at extracting water from soils, particularly as the soil water content decreases. This result can be achieved, for instance, by plants with deep roots and shallow growth angles (Lynch 2013). Such a root trait should allow crops to extract water stored in the subsoil more easily. Additionally, besides altering themselves, plants may also change their surroundings' properties to optimise access to water. Here, we study a plant trait that is expected to increase the ability of roots to extract water from dry soils: mucilage exudation.

Mucilage is a polymeric gel that is exuded by most plant roots. It is primarily composed of polysaccharides, but it contains also a small fraction of lipids (Read *et al.* 2003). Notably, fully hydrated mucilage has a water content of $10^5\%$

(the weight of fresh mucilage divided by its DW). McCully and Boyer (1997) discussed the potential role of mucilage and its large water-holding capacity in protecting roots from drought. They found that mucilage loses most of its water at water potentials higher than -0.01 MPa and concluded that mucilage water content, *per se* does not play a significant role in drought protection. Indeed, the volume of water held by mucilage when the soil is dry would be quickly consumed by plants.

To better understand the role of mucilage in plant–soil water relations, it is useful to look at the fluxes of water. Carminati *et al.* (2011) calculated the gradients in water potential in the soil near roots covered and not covered with mucilage. In this modelling study, they hypothesised that: (1) mucilage increases the water content in the rhizosphere at any water potential; (2) mucilage increases the viscosity of the soil solution and, consequently, it decreases the saturated conductivity of the rhizosphere; and (3) thanks to the increased water content, the rhizosphere has a higher unsaturated conductivity at low water potential. The simulations showed that mucilage facilitated the water uptake of roots. Thanks to mucilage, roots had a higher water potential during the drying period, wilted at a lower soil water potential and wilted later. Carminati and Vetterlein (2013) elaborated further on this result and suggested that the rhizosphere plays a key role in the acquisition of scarce resources. However, the role of mucilage and the rhizosphere on root water uptake and

drought tolerance has not yet been demonstrated and remains largely hypothetical.

In this manuscript, we aimed to experimentally test the hypothesis that mucilage exudation facilitates root water uptake from relatively dry soils. Since measuring water flow in soil–root systems is not trivial and the overall hydraulic conductance of the root–soil system depends on the properties of the roots, rhizosphere and soil, all being dynamic and hysteretic, we simplified the problem as follows: we substituted real roots with artificial ones – namely, we used suction cups of known hydraulic conductivity – and we used mucilage extracted from chia seeds (*Salvia hispanica* L.). There are two reasons for using chia seeds: (1) chia seeds produce large quantity of mucilage, and (2) the chemical composition of mucilage of chia seeds is similar to that of maize (*Zea mays* L.), both being mainly composed of xylose, glucose and uronic acids (25%). Furthermore, the mucilage from chia seeds has similar physical characteristics to maize and lupin (*Lupinus albus* L.) mucilage: it forms a gel when it is hydrated and it turns hydrophobic after drying.

The water flow into artificial roots covered with mucilage and placed into a sandy soil was measured with the root pressure probe technique (Liu *et al.* 2009). We employed the pressure probe (Der Werkzeugmacher, Eckersdorf, Germany) to monitor the pressure relaxation after imposing a given flow of water into or out of artificial roots covered and not covered with mucilage. The root pressure probe has been typically used to measure the hydraulic conductivity of excised roots in hydroponics. Here, we extended the technique to roots in relatively dry soils. The nonlinearity of the soil hydraulic conductivity as a function of the soil water content makes the interpretation of the relaxation experiment not trivial. Therefore, we developed a numerical model that simulates the water flow across soils mixed with mucilage. The novelty of the model consists in adding a nonequilibrium term to the Richards' equation that is usually used to simulate water flow in soils. The method is an extension of the nonequilibrium model of Carminati (2012).

We used this method to determine the hydraulic conductivity of the soil near the root when mucilage was or was not present. We aimed to test the hypotheses that mucilage facilitates water flow into the roots and that it increases the soil hydraulic conductivity. In a complementary experiment, we tested to what extent mucilage is able to hold water after its injection into a relatively dry soil.

Materials and methods

Mucilage preparation

We used mucilage from chia seeds (*Salvia hispanica* L.) as a model of plant mucilage. The chemical composition of mucilage of chia seeds is similar to that of maize (*Zea mays* L.): both are mainly composed of xylose, glucose and uronic acids. In both mucilages, the content of uronic acid is around 25% (Lin *et al.* 1994; Carminati and Vetterlein 2013). Furthermore, the mucilage of chia seeds showed a physical behaviour similar to that of mucilage from maize and lupins (*Lupinus albus* L.). It has been shown that the mucilage of chia seeds contains significant amounts of polysaccharides that form a gel-like

network around the seeds (Lin *et al.* 1994; Muñoz *et al.* 2012). Indeed, when we immersed chia seeds in water, mucilage appeared immediately as a transparent capsule around the seeds. We also dried the mucilage of chia seeds and observed a high contact angle, which is an indication of hydrophobicity.

Mucilage was extracted by the following procedure: we added 50 g of distilled water to 5 g of chia seeds. The mixture was stirred with a magnetic stirrer for 2 min and then kept for 2 h at room temperature. The mixture was then pushed through a sieve using a syringe that was cut at the end. In this way, we separated the seeds from their mucilage. It has to be mentioned that part of the mucilage, the most sticky and gel-like part, remained attached to the seeds.

Soil preparation

We used a sandy soil collected from the artificial catchment of Chicken Creek, located near Cottbus, Germany. The soil (sieved to a particle size smaller than 1 mm) consisted of 92% sand, 5% silt and 3% clay. We packed the soil homogeneously into aluminium containers measuring 6 cm × 10 cm × 1.5 cm. For the soil packing, we poured the soil through a sieve while the containers were laid horizontally to minimise soil layering. This procedure resulted in a soil with a bulk density of 1.45 g cm⁻³. We then put the samples on a balance and slowly added 2.5 g of water from the top into the soil. This resulted in an average soil water content of 0.03 cm³ cm⁻³.

Water retention curve and saturated hydraulic conductivity

We determined the water retention curve of the soil mixed with and without mucilage using a pressure plate apparatus (Soil moisture equipment, Santa Barbara, CA, USA.). We filled three cylinders 6 cm in diameter and 6 cm in height with dry soil. We compacted the soil slightly to achieve a density of 1.45 g cm⁻³ and saturated it with water. We placed the cylinders in a pressure plate apparatus and determined the soil water content at different levels of water potential. In order to determine the retention curve of the soil–mucilage mixture, we first mixed mucilage with the soil. Mucilage was extracted from the chia seeds as described above. After extraction, mucilage was well hydrated and had a concentration of 0.0125 g g⁻¹ (g of dry mucilage per g of water). We mixed 600 g of soil with 600 g of wet mucilage. We then incubated the mixture in the oven at 60°C for 24 h. Three cylinders were filled with the treated soil and the water content was measured at different water potential levels in the pressure plate apparatus. Note that to wet these samples, we had to leave them immersed in water for 5 days. We also measured the saturated hydraulic conductivity of the soil without mucilage by applying a constant pressure difference at the top and bottom of the cylinders, and measuring the outflow.

Drying rate of mucilage in soil

The objective of this experiment was to estimate how long and to what extent mucilage keeps the soil near the roots wet. We filled four containers of 9.5 cm × 9 cm × 8 cm with 500 g of dry soil. The packing resulted in a bulk density of 1.45 g cm⁻³ and a soil depth of 4 cm. We then added 14 g of water to the soil to reach a water content of 0.04 cm³ cm⁻³. We closed the lid of

each plastic box tightly and let the soil water content equilibrate for 10 h. We then injected 0.20 g of mucilage at seven points at regular intervals of ~3 cm.

After injection, we closed the containers with a plastic lid to avoid evaporation. We then collected soil samples at different time steps: 0 h, 3 h, 5 h, 7 h, 9 h, 12 h and 24 h after mucilage injection. The soil samples were collected using an auger with a diameter of 0.4 cm and a length of 7 cm. This experiment aimed to determine the water content of the soil regions embedded with mucilage over time. As a control experiment, we repeated the measurements but injecting water (using the same amount) instead of mucilage. We repeated these experiments in four replications.

Pressure probe experiment

We used a root pressure probe technique (Der Werkzeugmacher) to quantify the ease of the water flow into artificial roots covered and not with mucilage and placed into the soil. Root pressure probes have been used to measure the hydraulic conductivity of roots in hydroponics (Frensch and Steudle 1989; Ranathunge *et al.* 2004; Knipfer and Fricke 2010). Here, we applied the technique to artificial roots sitting in a relatively dry soil.

The root pressure probe consists of an oil-filled pressure chamber attached to a pressure transducer. The transducer converts the pressure to a voltage recorded on a computer. A metal rod is sealed inside the pressure chamber. The metal rod can be turned forward (or backward) to impose a flow into (or out) of the artificial root. The two directions induce injection or suction, respectively. A glass capillary with an internal diameter of 300 μm was attached to the pressure chamber and connected to the artificial root with the aid of silicon robber seals (Fig. 1). The first half of the capillary glass attached to the pressure chamber was filled with silicon oil and the half connected to the artificial roots was filled with degassed water. A meniscus formed at the oil–water interface that allowed us to control the volume of water moved during the injection and suction experiments.

We used an artificial root as a model system to mimic a plant root. This choice let us focus on the effects of mucilage, without

having the complexity of a real root with its possibly varying hydraulic conductivity. The artificial root was a rhizon soil moisture sampler (Eijkelkamp, Giesbeek, The Netherlands). It consisted of a hollow cylinder made of a hydrophilic porous polymer with a pore size of 0.15 μm , a metal rod in the middle of the cylinder for mechanical support and a teflon tube that was glued to the hollow nylon cylinder. The permeating part of the root was 0.2 cm in diameter and 4.5 cm in length.

The soil had initially an average water content of $0.03 \text{ cm}^3 \text{ cm}^{-3}$. A slight depression was formed on the soil surface and the artificial root was placed inside. We uniformly covered the artificial root with 0.5 g of wet mucilage. We added 0.4 cm of dry soil on the artificial root and we gently pressed the soil to ensure a good contact of the soil with the artificial root. We then covered the soil surface with a plastic foil to minimise evaporation. Evaporation was nevertheless monitored during the experiment by leaving the sample on the balance throughout the measurement period. We waited for 1–2 h until the pressure measured by the root pressure probe was constant. We refer to the measured pressure as the root pressure. We then moved a known volume of water into or out of the artificial root by turning the metal rod and monitored the root pressure relaxation. We applied several injection and suction pulses at different times for a period of 24 h. At the end of the experiment, we removed the artificial root from the soil and placed it in water to measure the half-time of its relaxation. We compared the half-time of the pressure relaxation in water at the end of the experiment with the one at the beginning to test if cavitation occurred during the measurements. We repeated this experiment in three replications.

As a control, we repeated the pressure probe experiment in the same set-up with the only difference being that instead of 0.5 g of mucilage, we added 0.5 g of water on the top of the root. We applied several suction and injection pulses at different times for a period of 5 h. We conducted this experiment in three replications.

The hydraulic conductivity of the artificial root was measured by immersing the artificial roots in water and applying a given suction or injection pulse. Determination of the hydraulic conductivity of the artificial root sitting in a nutrient solution

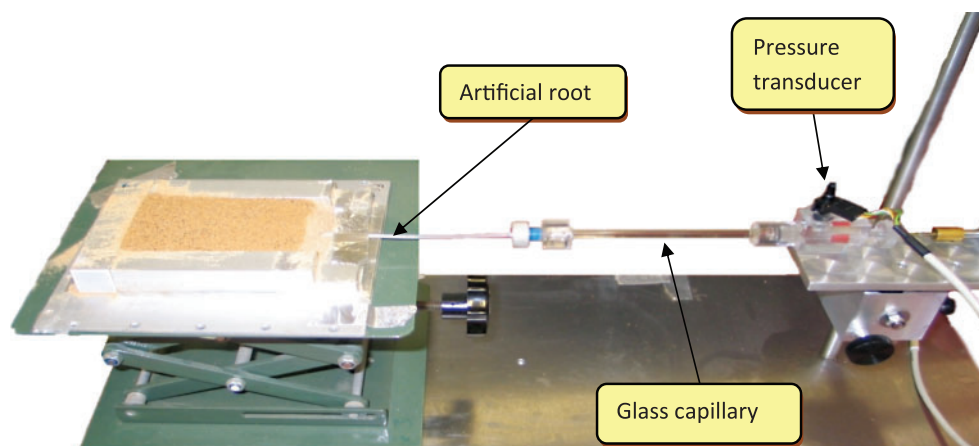


Fig. 1. Experimental set-up of the root pressure probe experiment.

is straightforward. In this case, the pressure relaxation follows an exponential curve and the half-time is inversely proportional to the hydraulic conductance of the system (Steudle *et al.* 1987).

The pressure relaxation curve in roots in soils is not exponential. Due to the movement of water through the soil and the nonlinearity of the soil's hydraulic properties, interpretations of the relaxation time requires numerical simulation of water flow in soils, with the additional complexity of mucilage dynamics. The model description is explained in the following section.

Model of water flow in roots and soil including mucilage dynamics

The experiments were simulated solving the Richards' equation of water flow in soils. When no mucilage was present in the soil, we assumed a unique function relating the volumetric water content (θ [$\text{cm}^3 \text{cm}^{-3}$]) to the soil matric potential (h), expressed here in pressure heads (cm); this relation is commonly called the soil water retention curve $\theta(h)$.

When mucilage was present in the soil, we assumed that the relation between θ and h was not stationary but that it changes over time. Carminati (2012) modelled the dynamic relationship between θ and h , introducing a nonequilibrium term in the Richards' equation. Here, we further developed this concept to enable a continuous transition of hydraulic properties from the root surface (where the concentration of mucilage is high) to the bulk soil (where the mucilage concentration decreases to zero).

The reason of the nonequilibrium between θ and h is that the gel-like properties of mucilage cause a time delay in the changes in water content. Since a certain fraction of the pore volume (R) may not be influenced by mucilage (because mucilage does not diffuse homogeneously throughout the pore space), the model of Carminati (2012) is modified as shown in Eqn 1:

$$\frac{\partial \theta}{\partial t} = RC_{\text{soil}}(h) \frac{\partial h}{\partial t} + (1 - R) \frac{1}{\tau} [h - h^{\text{eq}}(\theta_G)], \quad (1)$$

where $C_{\text{soil}} = \frac{d\theta_{\text{soil}}(h)}{dh}$ is the water-holding capacity of the bulk soil, R is the fraction of the pore volume that is not influenced by mucilage, τ describes the relaxation time of the nonequilibrium process, θ_G is the water content in the mucilage-filled pore space and $h^{\text{eq}}(\theta_G)$ is the matric potential that would be reached if the mucilage-filled pore space were kept at a steady water content (θ_G).

The parameter R in Eqn (1) is a weighting factor between the water-holding properties of the pores that are not filled with mucilage and the nonequilibrium water-holding dynamics of the mucilage-filled pores. The parameter τ determines the relaxation time of the process (e.g. a small τ (a short relaxation time) leads to a fast change in water content and fast equilibration).

In our simulation, the weighting factor R depends on the concentration of mucilage C_{tot} (the weight of dry mucilage per weight of dry soil) according to Eqn 2:

$$R(C_{\text{tot}}) = e^{-aC_{\text{tot}}}, \quad (2)$$

where a is a fitting parameter that depends on the properties of the soil and mucilage. Equation (2) guarantees that the hydraulic properties and nonequilibrium dynamics are not greatly affected

when mucilage concentrations are low. Due to diffusion of mucilage into the soil, it is reasonable to assume that the concentration of mucilage decreases continuously from a certain value at the root surface to zero at a certain distance from the root surface in the soil.

The relaxation time depends on the water content in the gel-filled pores:

$$\tau(\theta_G) = \theta_G^{-\gamma} \tau_0, \quad (3)$$

where γ and τ_0 are the fitting parameters. Equation (3) guarantees that the relaxation time is faster when soil and mucilage are wet.

Water holding capacity

The water retention curve of the soil was fitted with the Brooks–Corey parameterisation (Brooks and Corey 1964):

$$\Theta = \left(\frac{h}{h_0} \right)^{-\lambda}, \quad (4)$$

where Θ is the saturation. The saturation $\Theta = (\theta - \theta_r) (\theta_s - \theta_r)^{-1}$ depends on water content θ , residual water content θ_r and saturated water content θ_s . h_0 (cm) is the air-entry value and λ is a fitting parameter.

Mucilage binds to the soil particles and it is expected to increase the soil water content at negative water potentials (Albalasmeh and Ghezzehei 2014). Flory (1953) described the mucilage as a concentrated solution enclosed in a membrane. Actually, the membrane does not exist, but it represents the links between the polymers forming the solid network of the gel. The resulting osmotic forces depend on the concentration of mucilage in the water phase. The following equation (Eqn 5) has been used to fit the measured water retention of the mucilage samples:

$$h_{\text{gel}}(\theta) = h_{\text{soil}}(\theta) - \omega_0 \left(\frac{C_{\text{tot}}}{\theta} \right)^{\beta}, \quad (5)$$

where ω_0 and β are the fitting parameters and $h_{\text{soil}}(\theta)$ is the inverse of the water retention curve of the soil. The right term takes additional forces caused by the water-holding capacity of mucilage into account. Theoretically, the exponent β should be equal to 1. However, mucilage consists of long polymers forming a complex network and we considered β to be a fitting parameter.

As mucilage is exuded into the soil, its water potential tends to equilibrate with the one in the bulk soil. Depending on the initial water potential of the mucilage relative to the soil, mucilage will adsorb water from the soil or will lose water to it. Once the water potential of mucilage and bulk soil are in equilibrium, mucilage will not exert additional osmotic pressure on the root and will not limit root water uptake. This situation is different from the case of salts accumulating in the rhizosphere. In fact, salts do not easily enter the roots, although they can still diffuse in the soil, creating an osmotic potential difference between the root and the bulk soil. On the contrary, mucilage binds to the soil particles and does not freely diffuse into the soil (Albalasmeh and Ghezzehei 2014).

Hydraulic conductivity

The Mualem–Brooks–Corey model (Eqn 6) was used to describe the unsaturated hydraulic conductivity:

$$K(\theta) = K_{\text{sat}} \theta^{\frac{2}{1-L} + 2L}, \quad (6)$$

where L is tortuosity. The parameter L was estimated by fitting the root pressure relaxation in the experiments without mucilage.

The presence of mucilage in the liquid phase is expected to increase the viscosity of the soil solution. The increase of viscosity due to solutes is described with the following law (Eqn 7), modified from Morris *et al.* (1981):

$$\mu(c) = \mu_w(1 + vc^d), \quad (7)$$

where c is the concentration of mucilage in the liquid face, μ_w is the viscosity of water, and v and d are the fitting parameters.

The viscosity of the liquid phase leads to the following reduction in hydraulic conductivity:

$$K_G(\theta) = \frac{\mu_w}{\mu(c)} K(\theta). \quad (8)$$

Numerical solution

The experiments were simulated in one-dimensional radial geometry. The Richards' equation was solved according to Campbell (1985): the mass balance for each node was minimised with a Newton–Raphson method, where the Thomas algorithm was used in each step to solve the linearised equations.

Due to the high nonlinearity of the problem, an implicit Euler scheme was chosen for the time discretisation. Because of the nonequilibrium dynamics, water potential and water content were two independent variables and the water content had to be updated after each time step according to Eqn (1).

Initial and boundary conditions correspond to the values of the experiments. The parameters of the model have been hand-fitted in order to obtain a good correlation with the root pressure probe experiments and the mucilage drying experiment.

Results

Soil hydraulic properties

Fig. 2 shows the effect of mucilage on the soil water retention curve. The soil mixed with mucilage retained more water than the control soil at any given water potential. The increased water content of the soil–mucilage mixture is the result of the water adsorbing capacity of mucilage. In fact, Lin *et al.* (1994) and Muñoz *et al.* (2012) showed that the mucilage of chia seeds, upon wetting, formed a gel-like network with a remarkable capacity to adsorb water.

The water retention curve for the soil was fitted using the Brooks–Corey parameterisations (Eqn 4). The effect of mucilage was fitted using Eqn 5. The fitted parameters are shown in Table 1.

Drying rate of mucilage in soil

To determine how long mucilage keeps the soil near the root wet, we injected 0.2 g of mucilage in a soil with a water content of $0.04 \text{ cm}^3 \text{ cm}^{-3}$ and we measured the decrease in water content at the injection site. For comparison, we repeated the experiment by

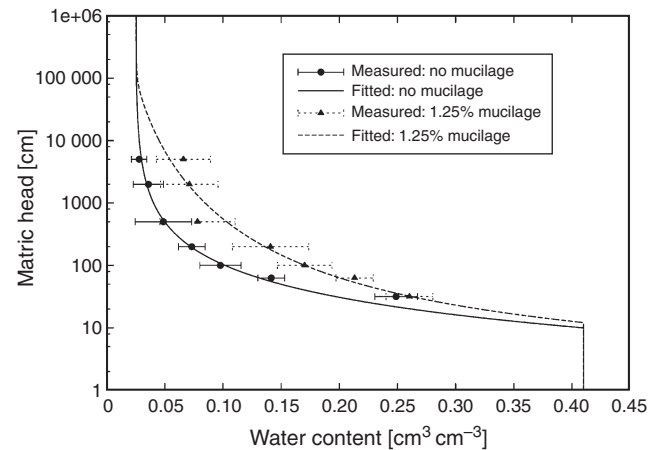


Fig. 2. Effect of mucilage on the soil water retention curve. The soil mixed with mucilage retained more water than the control soil at any given water potential. The data points are averages of three replications. The error bars show the s.d. The water retention curve of the soil without mucilage is fitted with the Brooks–Corey model (solid line). The data of the soil with mucilage are fitted with Eqn 5. The fitted parameters are presented in Table 1.

Table 1. Brooks–Corey parameters of the soil water retention curve

The parameters were obtained by fitting the measurements with pressure plate apparatus. The retention curve of the soil–mucilage mixture is parameterised according to Eqn 4. θ_s , saturated water content; θ_r , residual water content; λ , fitting parameter; h_0 , air entry value; β , fitting parameter

θ_s	θ_r	λ	h_0 (cm)	ω_0 (cm)	β
0.41	0.025	0.7	−10	1.22×10^6	3.80

injecting 0.2 g of water instead of mucilage. Fig. 3 shows the change in the volumetric water content after injection of mucilage and water. The results showed that the soil is drained much more slowly after mucilage injection than after water injection. One minute after injection, the soil water content was 2–3 times greater than it was after water injection. The samples injected with mucilage approached equilibrium at a water content of $0.11 \text{ cm}^3 \text{ cm}^{-3}$ in a period of ~24–48 h. The samples injected with water reached a local water content of $0.05 \text{ cm}^3 \text{ cm}^{-3}$ within 1–2 h after injection. The higher water content at equilibrium for the mucilage injection is in line with the results of Fig. 2.

The experiment with water injection was fitted solving the Richards' equation in radial coordinates. The experiment with mucilage exudation was fitted solving the Richards' equation modified according to Eqns 1–8. The slow drainage of the mucilage is explained by two factors: the increased viscosity of mucilage (and the consequent decrease in soil hydraulic conductivity) and the slow relaxation (the left-hand term of Eqn 1).

Pressure pulse experiment

Fig. 4 shows the pressure relaxation curves during several injection and suction pulses using artificial roots covered or

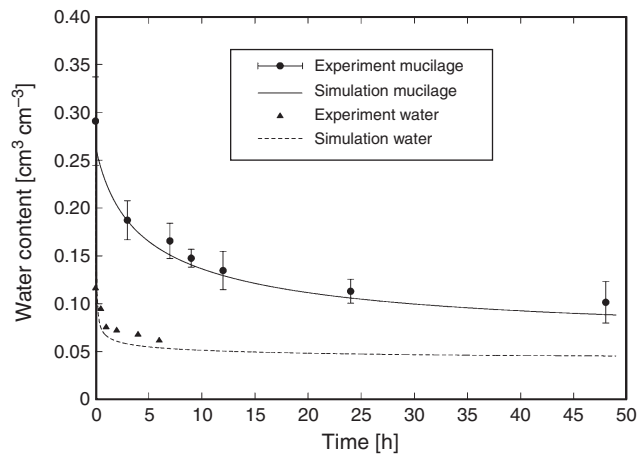


Fig. 3. Change in the volumetric water content after the injection of mucilage and water. The soil had an initial water content of $0.04 \text{ cm}^3 \text{ cm}^{-3}$. The figure shows that the soil was drained much more slowly after mucilage injection than after water injection. The experiments were fitted solving the modified Richards' equation according to Eqns (1–8). The parameters are presented in Table 2.

not covered with mucilage. During injection (positive pulses), the metal rod was quickly moved towards the artificial root. The water pressure inside the root suddenly increased and induced a flow of water out of the root into the soil. As the water exited the roots, the root pressure decreased until it reached the water pressure in the soil near the root. During suction (negative pulses), the metal rod was quickly withdrawn, reducing the water pressure inside the root and inducing a water flow from the soil into the root.

Fig. 4a shows the pressure curve during the experiments with the control soil (no mucilage around the artificial root). Before placing the artificial root into the soil, we measured the pressure relaxation in water. This gave the elasticity and the hydraulic conductivity of the artificial root. After placing the artificial root in soil, we covered it with an additional 0.4 cm of dry soil to ensure good contact between the root and soil. Due to the low soil matric potential of the newly added soil, the root pressure dropped. As the added soil grew wetter, the root pressure partly recovered (around 30 min after placing the root in the soil). We waited around 1 h until the pressure reached a constant value. We then applied a series of injection and suction pulses over a period of 5 h. The pressure relaxation in the suction experiment was slower than in the injection ones. This was caused by the decrease in water content near the root during the suction experiment and by the consequent reduction in hydraulic conductivity. We did not observe any significant change in pressure relaxation over time, indicating that the soil water content was close to equilibrium.

The curves were fitted by solving the Richards' equation in radial coordinates. The hydraulic conductance (K_{root}) and the elasticity (ϵ) of the root pressure probe system were obtained by fitting a suction experiment with the root sitting in free water. The height of the peak gave an elasticity of $\epsilon = 1.75 \times 10^5 \text{ cm cm}^{-3}$ and the relaxation time gave a root conductance of $K_{\text{root}} = 5.74 \times 10^{-7} \text{ cm}^2 \text{ s}^{-1}$ (value for the entire root surface).

The parameters of the soil water retention curve were estimated by fitting the values shown in Fig. 2 and Table 1. Additionally, we measured the saturated hydraulic conductivity of the soil $K_{\text{sat}} = 8.48 \times 10^{-3} \pm 0.13 \times 10^{-3} \text{ cm s}^{-1}$. The only parameter that was fitted was the tortuosity (L). According to the literature, an upper value of 2 is recommended for the L of sandy soil. However, simulations with $L=2$ did not match the observed curves. The bad fit was probably caused by partial contact between the artificial root and sand. Indeed, during the measurements, we realised how difficult was to place the root in uniform contact with the soil without leaving any gap. We included such complexity by introducing a thin layer of 1 mm thickness at the root surface, where we set a different L . We obtained the best fit using $L_{\text{bulk}} = 2$ for the bulk soil and $L_{\text{int}} = 4.8$ for the interface layer. This value is similar to the L of 5 obtained by Carminati *et al.* (2008) to describe the similar problem of partial contacts between soil aggregates. It is important to mention that the fitting was very sensitive to the properties of the soil near the roots.

Fig. 4b–d shows the pressure curves during one experiment with an artificial root covered with mucilage and placed into the soil as described above. Compared with the experiments without mucilage, the drop in root pressure after placing the root in the soil was much smaller and the pressure reached equilibrium faster. This was because the whole surface of the artificial root was covered with mucilage that had a higher water potential than the soil. However, the root pressure at equilibrium was similar in the two experiments, suggesting that even when covered with mucilage, the root equilibrates with the water potential of the bulk soil. When the root pressure became constant, we applied a series of injection and suction pulses over a period of 24 h. The pressure relaxation in roots covered with mucilage was much faster than it was in roots that were not covered with mucilage, particularly in the suction experiment. A quicker pressure relaxation indicates an easier flow of water across the soil–root interface. This proves that the water flow into the root was facilitated by mucilage.

Over time, the pressure relaxation became slower (Fig. 5). Fig. 5 shows the suction pulses from different samples at increasing time. After mucilage application, the pressure relaxation was quicker than in the soil without mucilage. After 24 h, the relaxation times became similar. This was probably caused by the drainage of mucilage over time, as shown in Fig. 3.

The pressure relaxation curves were fitted solving the Richards' equation modified according to Eqns 1–8. The fitting parameters are shown in Table 2. The model was capable of reproducing the evolution over time of the pressure relaxations well. The fast pressure relaxation in the root covered with mucilage was caused by the high hydraulic conductivity of the soil–mucilage mixture next to the root. The evolution of the hydraulic conductivity in the soil next to the root for both cases with and without mucilage is plotted in Fig. 6. Fig. 6 shows that in both cases, the hydraulic conductivity decreased after placing the root in soil. This was caused by the drying of the soil near the root surface. Mucilage increased the hydraulic conductivity of the soil next to the root for the measured 24 h. During the first 5 h, the conductivity was increased by a factor of 10. After 24 h, the conductivity was increased by a factor of two. The decreasing difference over time was caused by mucilage drying.

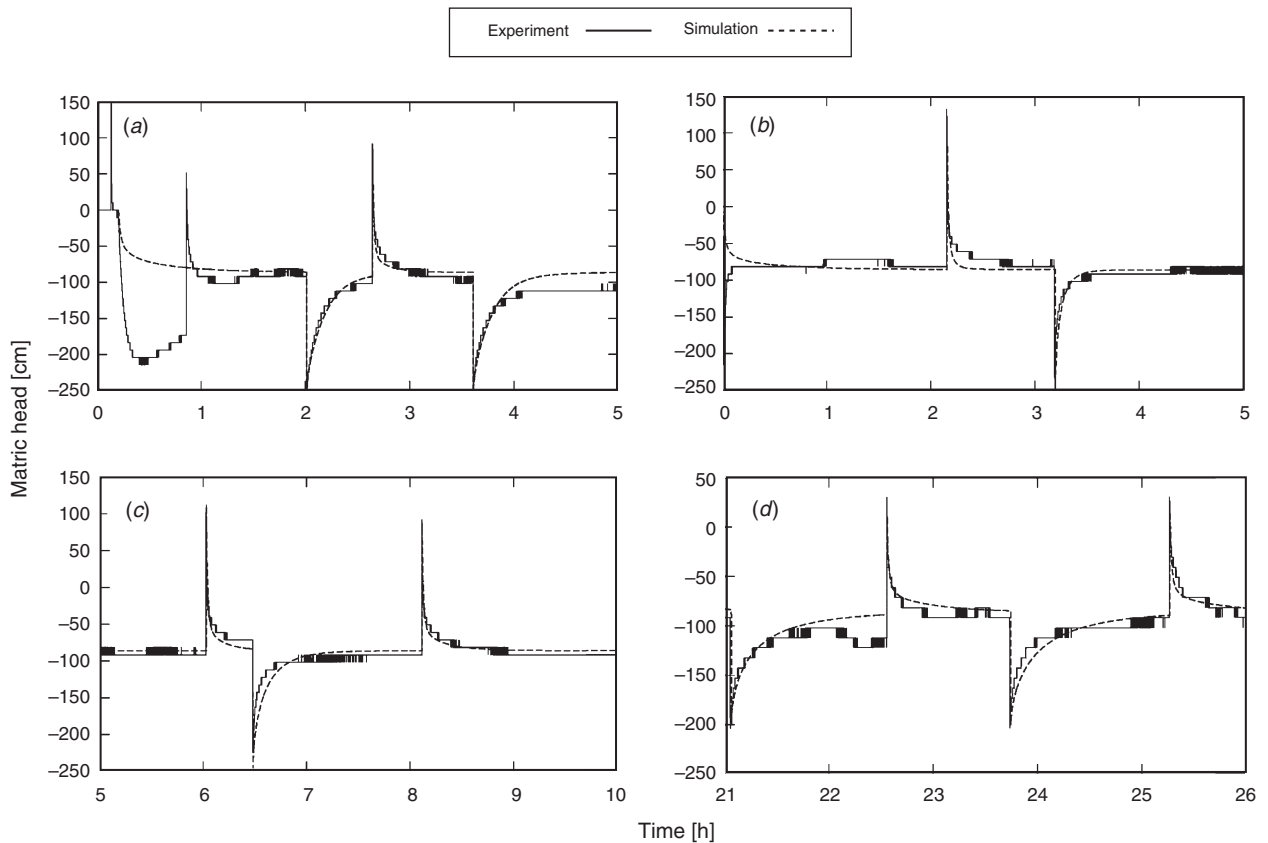


Fig. 4. Pressure relaxation curves during several injection and suction pulses with artificial roots covered and not covered with mucilage. (a) Pressure curve for a root not covered with mucilage. After we placed the artificial roots in soil, we covered them with additional dry soil to achieve a good contact between root and soil. Due to the high water tension in the newly added soil, the root pressure dropped. When the pressure reached a constant value, we applied a series of injection and suction pulses over a period of 5 h. The curves were fitted by solving the Richards' equation in radial coordinates. Pressure curves for an artificial root covered with mucilage over (b) 0–5 h, (c) 5–10 h and (d) 21–26 h (note the different scale in the y axis in (d)). The drop in root pressure after placing the root in the soil was smaller and the pressure reached equilibrium faster. The pressure relaxations in roots covered with mucilage were initially much faster and then they became slower over time. The pressure relaxation curves were fitted solving Richards' equation modified according to Eqns (1–8). The fitting parameters are shown in Table 2.

Interestingly, the model was able to explain the following apparently contradicting results: the soil mixed with mucilage was drained much more slowly than the pure soil (Fig. 3), which seems to suggest a lower hydraulic conductivity of the soil mixed with mucilage. On the other hand, the pressure pulse experiments show that mucilage allowed water to flow more easily through a dry soil. These results would be difficult to model with the classical Richards' equation. On the contrary, the nonequilibrium model (Eqn 1) is able to reproduce both observations. Indeed, the two observations are not in contradiction. The slow drainage of the gel is caused by the slow diffusion of mucilage (and its solid network) through the pore space as well as by the slow drying of mucilage and the consequent shrinking of its solid network. This was modelled by introducing the nonequilibrium term on the left side of Eqn 1. The consequent high water content in the soil–mucilage mixture results in an increase in the hydraulic conductivity of the soil near the root. In summary, the easier water flow to the root due to mucilage is the result of the slow spreading and drying of mucilage and the consequent increase in hydraulic conductivity.

Only two of the parameters used for the simulation of the pulse experiments (Fig. 4*b–d*) were different from those used for simulating the experiment in Fig. 3. This was probably caused by the different way we injected mucilage in the two experiments. In the pulse experiments, we placed mucilage around the root and then we added dry soil on the top. This may have resulted in a preferential distribution of mucilage on the lateral sides of the root, which cannot be captured in our radial model. On the other hand, in the experiment of Fig. 3, we simply injected mucilage in the soil. The two procedures may have resulted in different local soil densities as well as different spatial distributions of the gel.

Discussion

Our simple experiment showed that mucilage increased the unsaturated hydraulic conductivity of dry soils. Mucilage exudation is therefore expected to favour root water uptake in relatively dry soils, when the hydraulic conductivity of the soil near the roots is expected to drop by several orders of magnitude and become the major resistance to the water flow into roots. By increasing the hydraulic conductance of the rhizosphere,

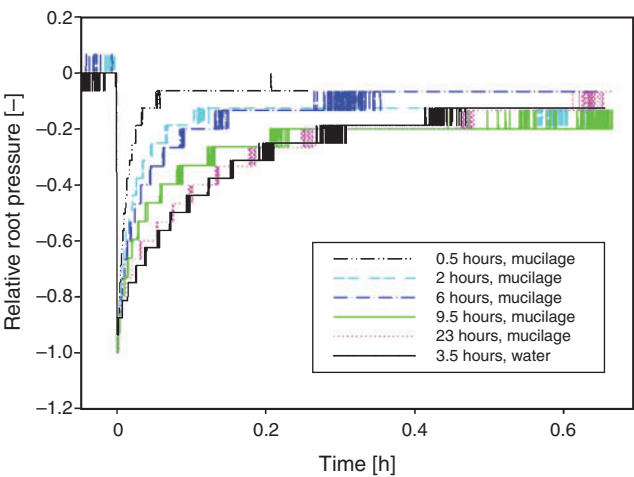


Fig. 5. Pressure relaxation curves with artificial roots covered and not covered with mucilage at different times. After mucilage application, the pressure relaxation was much quicker than in the soil without mucilage. After 24 h, the relaxation times became similar. The data are normalised between 0 and 1 for comparison.

Table 2. Parameters of the nonequilibrium model (Eqn 1)
The data are obtained from fitting the root pressure probe measurements and the drying of mucilage

Experiment	<i>a</i>	γ	τ_0 (s cm ⁻¹)	ν	<i>d</i>
Root pressure probe	100	4	1.43×10^4	566	1.4
Injection of mucilage	333	4	1.67×10^5	566	1.4

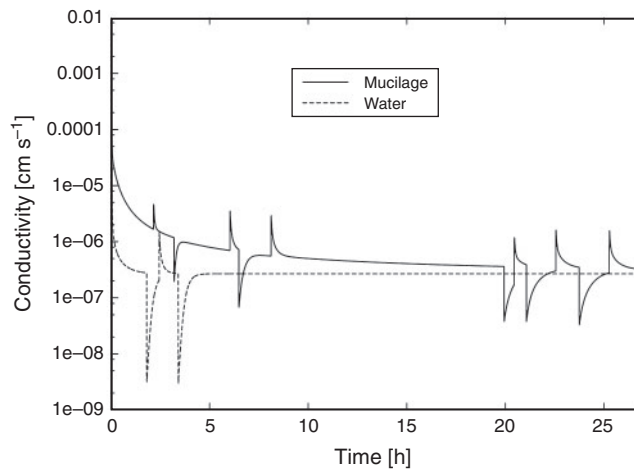


Fig. 6. Evolution of the hydraulic conductivity in the soil next to the root for both cases, with and without mucilage. The hydraulic conductivity decreased after placing the root in soil. Mucilage increased the hydraulic conductivity of the soil next to the root for the measured 24 h. During the first 5 h, the conductivity was increased by a factor of 10. After 24 h, the conductivity increased by a factor of 2.

mucilage can help plants to better capture water under dry conditions. The increase in hydraulic conductivity resulted from the capacity of mucilage to stay well hydrated.

Specifically, we observed that: (1) mucilage increased the equilibrium water content of soil at any water potential (Fig. 2); and (2) mucilage did not spread quickly into the soil and it was slowly drained (Fig. 3). Both results confirm the hypothesis of Young (1995) that mucilage increases the water content of the rhizosphere compared with that of the bulk soil. The results do not contradict the measurements of McCully and Boyer (1997), who showed that mucilage loses most of its water at high water potentials. The remaining volume of water retained by mucilage at more negative water potentials accounts for the increase in water content of $\sim 0.05 \text{ cm}^3 \text{ cm}^{-3}$ (Fig. 2).

Our experiments were carried out using artificial roots and mucilage collected from chia seeds. This choice was functional to perform well-controlled experiments, avoiding the complexity of variable root conductivity (Steudle 2000; Maurel *et al.* 2008). Additionally, it can be argued that mucilage from chia seeds has different properties than mucilage exuded by plant roots in the field. However, following similar reasoning, we argue that it is unlikely that mucilage properties are equal in all plant species, soil types and environmental conditions. We observed that mucilage from chia seeds had a similar behaviour as that from maize and lupine: it formed a gel when immersed in water and it turned hydrophobic after drying. A similar chemical composition was also reported (Lin *et al.* 1994). Our conclusions on the effect of mucilage on root water uptake can therefore be generalised to more realistic cases, although the extent of these effects will vary among plants, soils and environmental conditions. For instance, we carried out our experiments in sandy soils. We expect that in more loamy and clayey soils, the effects of mucilage on the soil hydraulic conductivity would be smaller.

We showed that at a water content of $0.03\text{--}0.04 \text{ cm}^3 \text{ cm}^{-3}$, mucilage kept the soil wet and well conductive for a period of 24 h. Afterwards, mucilage was drained and its hydraulic conductivity approached that of the original soil. The effects of mucilage are therefore highly dynamic and are expected to be more significant in the young root segments, whereas old root segments are likely to be covered with old and drained mucilage. Besides being drained, mucilage can also be degraded by microorganisms, which themselves are covered by extracellular polymeric substances (Chenu and Roberson 1996; Or *et al.* 2007). Drying, degradation and alteration of mucilage will therefore result in varying hydraulic properties of the rhizosphere along the root system. This conclusion follows the point of view of Watt *et al.* (2006), who reported that rhizosphere properties change as a function of distance from root tips and growth rate. Similarly, our findings support the hypothesis of Carminati and Vetterlein (2013), who suggested that rhizosphere plasticity helps plants to adapt better to the heterogeneous distribution of soil resources like water and nutrients. Our results are also in line with former observations of the wet rhizosphere around young root segments (Watt *et al.* 1994).

The effect of mucilage exudation depends on the capacity of the root system to transport water efficiently from the young root segments covered with well-hydrated mucilage. Assuming that roots grow at a velocity of 2 cm day^{-1} , we expect that mucilage facilitates the water flow into the first 2 cm near the root tip. Our study suggests that mucilage has positive effects on root water uptake, and these increase when roots grow faster and

when a larger portion of roots is covered with fresh and wet mucilage.

In conclusion, we conducted a simplified experiment to test the effects of mucilage on water flow in dry soils. The results demonstrated that mucilage facilitated the water flow in dry soils and support our hypothesis that mucilage exudation helps plant roots to take up water from dry soils. Our study is only a small initial step towards a better understanding of the role played by the rhizosphere in soil–plant water relations. Other experiments are necessary to estimate the effects of mucilage exudation in more realistic scenarios and how these effects vary among plant species, soil types and growing conditions. Possibly, these studies should also consider the effects on nutrient uptake and on microorganisms living in the rhizosphere. At the moment, what we can conclude is that mucilage has the potential to increase the capability of young root segments to capture water from soils, particularly when the soil is drying. Such characteristics potentially help plants to use soil resources and survive drought spells. A better understanding of mucilage and rhizosphere interactions may therefore bring new strategies to decrease drought stress in plants.

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